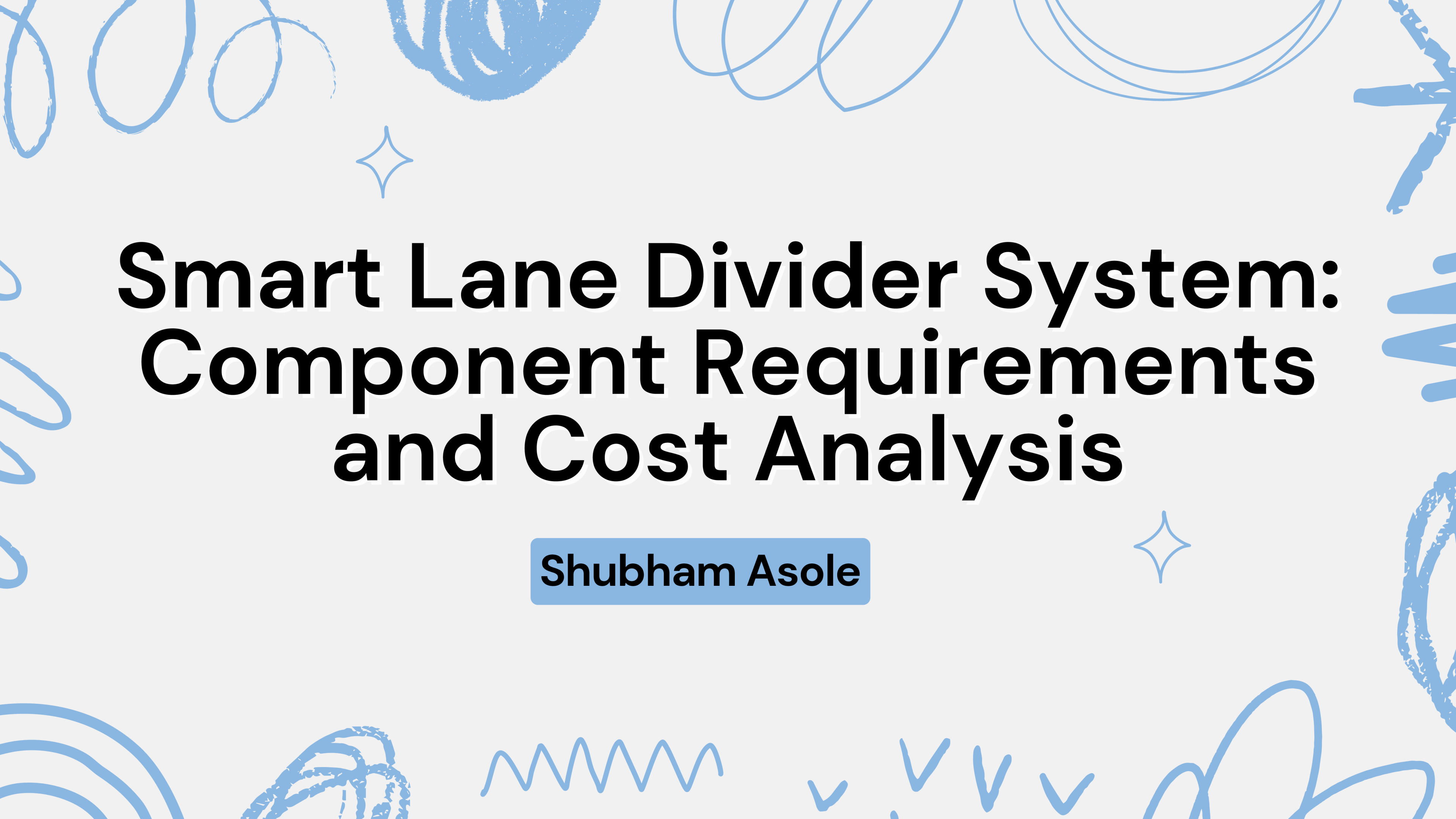




Smart Lane Divider System: Component Requirements and Cost Analysis

Shubham Asole

Origin of the creative idea

The concept of the smart fordable lane divider system was developed by Nitish.

Inspired by the limitations of traditional zipper truck systems and the need for faster, safer, and more adaptive traffic management, they proposed a dynamic lane reversal mechanism using FRP panels with steel-reinforced edges.

Their design integrates automation, structural innovation, and real-time control to optimize traffic flow and improve road safety



Raspberry Pi 4 (with Camera Module)

Material Requirement: Raspberry Pi 4 Model B (4GB) + Pi Camera Module

Cost: ₹ 5,549.00 + 3,574.00

Capital Cost: Medium – One-time investment for control and AI processing

Pros:

- Supports AI (OpenCV, YOLO)
- GPIO for sensor/motor control
- Portable & energy-efficient

Cons:

- Not industrial-grade
- Limited real-time processing power under heavy load

DC Gear Motor (High Torque)

Material Requirement: 12V/24V DC Gear Motor (30–60 RPM, high torque)

Cost: ₹1,000 – ₹2,000 each (2 motors = ₹2,000 – ₹4,000)

Capital Cost: Medium – Needed for panel movement and folding

Pros:

- Strong torque for sliding heavy panels
- Easily controllable

Cons:

- Requires precise speed control
- Slightly bulky

Rotary Encoder (Industrial Grade)

Material Requirement: Pro-Range 600 PPR 2-Phase Incremental Optical Rotary Encoder

Cost: ₹ 1,416

Capital Cost: Medium – One-time for position tracking accuracy

Pros:

- High resolution and accuracy
- Industrial durability

Cons:

- Requires extra programming and integration effort
- May be affected by mechanical slippage

Limit Switches (Heavy Duty)

Material Requirement: Snap-action limit switches

Cost: ₹439.58

Capital Cost: Low – Inexpensive safety components

Pros:

- Simple and reliable position feedback
- Works well in harsh environments

Cons:

- Only binary detection (no analog feedback)
- Needs mechanical contact to trigger

Ultrasonic Sensor (JSN-SR04T or HC-SR04)

Material Requirement: Waterproof JSN-SR04T or standard HC-SR04

Cost: ₹252

Capital Cost: Low – Optional safety enhancement

Pros:

- Detects objects up to 4–6 meters
- Easy to interface with Raspberry Pi

Cons:

- Affected by temperature and angle
- Slower than Lidar for rapid motion detection

Optional Lidar Sensor (TFmini-S)

Material Requirement: TFmini-S Lidar (range up to 12 meters)

Cost: ₹3,245

Capital Cost: Medium – Used only if high-precision obstacle detection is required

Pros:

- Higher accuracy and range than ultrasonic
- Fast response time

Cons:

- Higher cost
- Requires UART/I2C integration

Motor Driver (BTS7960 or L298N)

Material Requirement: High-current motor driver for gear motors

Cost: ₹249

Capital Cost: Low – Essential for motor control

Pros:

- Handles high current
- Compatible with Raspberry Pi

Cons:

- Heat dissipation required
- Wiring must be precise

Battery Pack

Material Requirement: Rechargeable sealed lead-acid or Li-ion battery

Cost:

Capital Cost: Medium – Critical for portable/independent operation

Pros:

- Powers motors and Pi
- Rechargeable

Cons:

- Heat dissipation required
- Wiring must be precise

Solar Panel + Charge Controller

Material Requirement: 50W–100W Solar Panel with 12V controller

Cost: ₹2,000 – ₹4,000 + ₹500 – ₹1,000 (controller)

Capital Cost: Medium – Eco-friendly solution for long-term use

Pros:

- Sustainable and low maintenance
- Reduces energy cost

Cons:

- Performance depends on sunlight
- Higher upfront cost than direct power

Summary of Capital Cost and Total Estimate

Material Cost Range: ₹20,000 – ₹30,000

Initial Capital Investment:

- One-time setup: Medium
- Maintenance cost: Low
- Operation cost: Minimal (solar + battery)

Total Investment Justification:

- Long-term sustainability
- Reduces need for manual lane redirection
- Improves traffic flow and safety



The background is a light gray color, decorated with various hand-drawn blue doodles. These include several loops and swirls at the top, a series of vertical lines on the right side, a wavy line at the bottom center, and several checkmarks at the bottom right. The central text is in a bold, black, sans-serif font with a white drop shadow.

**Thank you
very much!**